

K_5 MINUS AN EDGE IS 2-CONFORMALLY RIGID BUT NOT 1-CONFORMALLY RIGID

ABSTRACT. Let G be a graph on n vertices with edge set E , let $\Delta_E = \{w \in \mathbb{R}_{\geq 0}^E : \sum_e w_e = |E|\}$, and index the eigenvalues of the weighted Laplacian $L(w)$ increasingly as $0 = \lambda_1(w) \leq \dots \leq \lambda_n(w)$. Write $S_k(w)$ for the sum of the k largest and $s_k(w)$ for the sum of the k smallest nontrivial eigenvalues. Following Assumpção, Coutinho and Godsil, G is *k-conformally rigid* if the all-ones weight simultaneously minimises S_k and maximises s_k on Δ_E . In the Conclusion of their paper they ask whether there is a graph that is 2-conformally rigid but not 1-conformally rigid. There is one at $n = 5$: K_5 minus an edge. We answer the question with an exact rational dual certificate, a single 5×5 integer matrix, that proves $S_2(w) \geq 10 = S_2(\mathbf{1})$ for every $w \in \Delta_E$; since $n - 1 - k = 2 = k$ here, 2-conformal rigidity follows. At this order the duality is self-paired, so the two conditions defining 2-conformal rigidity coincide and one inequality discharges both; a witness at larger n , where the two conditions are genuinely distinct, is not provided here. That $K_5 - e$ is not 1-conformally rigid is already recorded as numerical data by Niu; we exhibit the exact rational weight behind it.

1. THE QUESTION, AND THE ANSWER

Let $G = (V, E)$ be a graph on n vertices. For $w \in \mathbb{R}_{\geq 0}^E$ let

$$L(w) = \sum_{\{a,b\} \in E} w_{ab} (e_a - e_b)(e_a - e_b)^\top$$

be the weighted Laplacian, and let its eigenvalues be indexed *increasingly*, $0 = \lambda_1(w) \leq \lambda_2(w) \leq \dots \leq \lambda_n(w)$. Put

$$\Delta_E = \left\{ w \in \mathbb{R}_{\geq 0}^E : \sum_{e \in E} w_e = |E| \right\},$$

$$S_k(w) = \sum_{j=n-k+1}^n \lambda_j(w), \quad s_k(w) = \sum_{j=2}^{k+1} \lambda_j(w),$$

so that S_k and s_k sum the k largest and the k smallest *nontrivial* eigenvalues. The all-ones weight $\mathbf{1}$ lies in Δ_E and $L(\mathbf{1})$ is the ordinary Laplacian. Following the definition of k -conformal rigidity in Assumpção, Coutinho and Godsil [1], call G *upper-k rigid* if $\mathbf{1}$ minimises S_k over Δ_E , *lower-k rigid* if $\mathbf{1}$ maximises s_k over Δ_E , and *k-conformally rigid* if both hold, for $k \in \{1, \dots, n-1\}$. For $k = 1$ this is the notion of Steinerberger and Thomas

2020 *Mathematics Subject Classification.* 05C50, 05C22, 90C22, 15A42.

Key words and phrases. conformal rigidity, weighted Laplacian, eigenvalue sums, Ky Fan principle, dual certificate, counterexample.

[4]: $\mathbf{1}$ minimises λ_n and maximises λ_2 ; see also Gouveia, Steinerberger and Thomas [5] for the $k = 1$ theory.

The Conclusion of [1] asks, in running prose,

“Are there graphs that are 2-conformally rigid but not 1-conformally rigid?”

The answer is yes, at $n = 5$.

Theorem 1. *Let $H = K_5 - e$ be the complete graph on $V = \{1, 2, 3, 4, 5\}$ with the pair $\{4, 5\}$ deleted, so $|E(H)| = 9$. Then H is 2-conformally rigid and is not 1-conformally rigid.*

Both halves are certified by objects printed below in full.

The negative half of Theorem 1 is not new as a fact: Niu’s dataset [3] tabulates lower- and upper-1 rigidity for the graphs of the House of Graphs census, and its row `id=450`, the 5-vertex 9-edge graph H , records `lcr=False`, `ucr=True`. What is supplied here is the dual certificate of Proposition 2, and the exact rational weight w^* of Proposition 4 behind that table cell. No claim of priority is made for the upper-2 fact itself: whether it already follows from results or data in [1] or [3] has not been checked here. Note that [3] calls a graph conformally rigid when it is lower-1 *or* upper-1 rigid, while [4] and [1] use *and*; we use the conjunction of [1] throughout.

2. THE TWO CERTIFICATES

Both proofs use the adjoint of $w \mapsto L(w)$. For a symmetric X put

$$(1) \quad \begin{aligned} L^*(X)_{ab} &= X_{aa} + X_{bb} - 2X_{ab} \quad (\{a, b\} \in E), \\ \text{so that} \quad \text{tr}(L(w)X) &= \sum_{e \in E} w_e L^*(X)_e, \end{aligned}$$

which is immediate from $\text{tr}((e_a - e_b)(e_a - e_b)^\top X) = (e_a - e_b)^\top X(e_a - e_b)$.

The spectrum of H is exact and elementary. Since $L(\mathbf{1}) = 5I - J - dd^\top$ with J the all-ones matrix and $d = e_4 - e_5$, a vector x with $x^\top \mathbf{1} = 0$ and $x_4 = x_5$ satisfies $L(\mathbf{1})x = 5x$, while $L(\mathbf{1})d = 3d$. Hence

$$(2) \quad \text{spec } L(\mathbf{1}) = \{0, 3, 5, 5, 5\}, \quad S_1 = 5, \quad S_2 = 10, \quad S_3 = 15, \quad S_4 = 18, \quad s_1 = 3,$$

the 5-eigenspace being $W = \{x : x^\top \mathbf{1} = 0, x_4 = x_5\}$, spanned by $a = (1, -1, 0, 0, 0)$, $b = (0, 1, -1, 0, 0)$ and $c = (2, 2, 2, -3, -3)$.

Proposition 2 (upper-2 rigidity). *Let*

$$(3) \quad X = \frac{1}{45}M, \quad M = \begin{pmatrix} 22 & -3 & -3 & -8 & -8 \\ -3 & 22 & -3 & -8 & -8 \\ -3 & -3 & 22 & -8 & -8 \\ -8 & -8 & -8 & 12 & 12 \\ -8 & -8 & -8 & 12 & 12 \end{pmatrix}.$$

Then $S_2(w) \geq 10 = S_2(\mathbf{1})$ for every $w \in \Delta_E$, so H is upper-2 rigid.

Proof. Four facts about X , each a direct computation with (3).

(1) X is symmetric with $\text{tr } X = (3 \cdot 22 + 2 \cdot 12)/45 = 90/45 = 2$.

(2) $X\mathbf{1} = 0$ and $Xd = 0$, and $Xa = \frac{5}{9}a$, $Xb = \frac{5}{9}b$, $Xc = \frac{8}{9}c$. The first two hold because every row of M sums to 0 and its last two columns are equal. For the others, $Ma = (25, -25, 0, 0, 0)$, which is $25a$, and $Mc = (80, 80, 80, -120, -120)$, which is $40c$; and $Mb = 25b$ because swapping the coordinates 1 and 3 permutes M to itself and carries a to $-b$. Since $\mathbf{1}, d, a, b, c$ are linearly independent (the determinant of the matrix with these columns is -30), the spectrum of X is exactly $\{0, 0, \frac{5}{9}, \frac{5}{9}, \frac{8}{9}\}$; equivalently the characteristic polynomial of X is $x^2(x - \frac{8}{9})(x - \frac{5}{9})^2$. In particular

$$0 \preceq X \preceq I$$

holds *exactly*, not to a tolerance.

(3) $L(\mathbf{1})X = 5X$, hence $\text{tr}(L(\mathbf{1})X) = 5 \text{tr } X = 10 = S_2(\mathbf{1})$. Indeed the range of X is spanned by a, b, c and lies in the 5-eigenspace W .

(4) $L^*(X)$ is the constant $\frac{10}{9}$ on all nine edges. For an edge inside $\{1, 2, 3\}$, $L^*(X) = \frac{1}{45}(22 + 22 + 6) = \frac{50}{45} = \frac{10}{9}$; for an edge from $\{1, 2, 3\}$ to $\{4, 5\}$, $L^*(X) = \frac{1}{45}(22 + 12 + 16) = \frac{50}{45} = \frac{10}{9}$. These are the only two edge types, so $\sum_{e \in E} L^*(X)_e = 9 \cdot \frac{10}{9} = 10$.

Now let $w \in \Delta_E$. Because $X\mathbf{1} = 0$, X acts on the $(n-1)$ -dimensional space $\mathbf{1}^\perp$, on which $L(w)$ has eigenvalues $\lambda_2(w), \dots, \lambda_n(w)$. The Ky Fan maximum principle [2] applied on that subspace gives

$$\text{tr}(L(w)X) \leq S_2(w) \quad \text{whenever } 0 \preceq X \preceq I, \text{tr } X = 2, X\mathbf{1} = 0.$$

By (1), (2), (1) and (4),

$$\text{tr}(L(w)X) = \sum_{e \in E} w_e L^*(X)_e = \frac{10}{9} \sum_{e \in E} w_e = \frac{10}{9} |E| = 10,$$

the constancy of $L^*(X)$ being exactly what makes this independent of w . Hence $S_2(w) \geq 10$ for every $w \in \Delta_E$, and by (3) equality holds at $w = \mathbf{1}$. $\square$

Lemma 3 (duality). *For $1 \leq k \leq n-2$ and every $w \in \Delta_E$ we have $S_k(w) + s_{n-1-k}(w) = 2|E|$. Consequently G is upper- k rigid if and only if it is lower- $(n-1-k)$ rigid.*

Proof. $\sum_{j=2}^n \lambda_j(w) = \text{tr } L(w) = 2 \sum_e w_e = 2|E|$ on Δ_E , and S_k and s_{n-1-k} partition that sum. A constant sum turns minimising one summand into maximising the other. $\square$

This is the duality of [1], stated there for the same range $k \in \{1, \dots, n-2\}$, which is what s_{n-1-k} requires.

Proposition 4 (failure of lower-1 rigidity). *Let $w^* \in \mathbb{R}_{\geq 0}^E$ take the value $\frac{9}{7}$ on the six edges meeting $\{4, 5\}$ and $\frac{3}{7}$ on the three edges inside $\{1, 2, 3\}$. Then $w^* \in \Delta_E$,*

$$\text{spec } L(w^*) = \left\{0, \frac{27}{7}, \frac{27}{7}, \frac{27}{7}, \frac{45}{7}\right\},$$

and $s_1(w^*) = \frac{27}{7} > 3 = s_1(\mathbf{1})$. Hence H is not lower-1 rigid, so it is not 1-conformally rigid; equivalently, $S_3(w^*) = \frac{99}{7} < 15 = S_3(\mathbf{1})$, so H is not upper-3 rigid either.

Proof. w^* is nonnegative and $6 \cdot \frac{9}{7} + 3 \cdot \frac{3}{7} = \frac{54+9}{7} = 9 = |E|$, so $w^* \in \Delta_E$. Write $\alpha = \frac{3}{7}$, $\beta = \frac{9}{7}$. With a, b, c, d as above: $L(w^*)a = (3\alpha + 2\beta)a = \frac{9+18}{7}a = \frac{27}{7}a$ and likewise for b ; $L(w^*)d = 3\beta d = \frac{27}{7}d$, because d is supported on $\{4, 5\}$, which is a non-edge, and each of 4, 5 has weighted degree 3β ; and reading off the first and fourth coordinates of $L(w^*)c$ gives $\frac{90}{7} = \frac{45}{7} \cdot 2$ and $-\frac{135}{7} = \frac{45}{7} \cdot (-3)$, so $L(w^*)c = \frac{45}{7}c$. Together with $L(w^*)\mathbf{1} = 0$ these five independent vectors give the stated spectrum, whose sum is $\frac{81+45}{7} = 18 = 2|E|$ as it must be. Now $s_1(w^*) = \lambda_2(w^*) = \frac{27}{7} > 3 = \lambda_2(\mathbf{1}) = s_1(\mathbf{1})$, so $\mathbf{1}$ does not maximise s_1 on Δ_E . The upper-3 statement is the same fact through Lemma 3 with $n = 5$, $k = 3$: $S_3(w^*) = \frac{27+27+45}{7} = \frac{99}{7}$ and $18 - \frac{27}{7} = \frac{99}{7} < 15$. $\square$

Proof of Theorem 1. For $n = 5$ and $k = 2$ we have $n - 1 - k = 2 = k$, so Lemma 3 turns upper-2 rigidity into lower-2 rigidity. Proposition 2 gives the former, hence both, hence H is 2-conformally rigid. Proposition 4 gives that H is not lower-1 rigid, hence not 1-conformally rigid. $\square$

Remark. At $n = 5$ and $k = 2$ the duality of Lemma 3 is self-paired, so the two conditions making up 2-conformal rigidity coincide and one inequality discharges both. In general they are distinct, and a witness at larger n would have to verify each; this note says nothing about that case.

3. VERIFICATION

Everything above is determined by the labelled graph H and the two printed objects M and w^* . The nine edges of H in the labelling of this note are

$$\{1, 2\}, \{1, 3\}, \{1, 4\}, \{1, 5\}, \{2, 3\}, \{2, 4\}, \{2, 5\}, \{3, 4\}, \{3, 5\},$$

and a machine-readable encoding is the graph6 string $D^{\wedge}\{$, on the vertex set $\{0, 1, 2, 3, 4\}$ with the pair $\{0, 1\}$ deleted, carried onto the labelling above by $0 \mapsto 4$, $1 \mapsto 5$, $2 \mapsto 1$, $3 \mapsto 2$, $4 \mapsto 3$.

The proofs are 5×5 rational arithmetic and can be checked by hand. A stand-alone program `verify.py` accompanies this note; it reads the graph6 string, the edge list, the matrix M and the weight w^* exactly as printed here, and re-derives in exact integer and `Fraction` arithmetic, using only the Python standard library and with no floating-point comparison anywhere: the graph and its edge list under the printed relabelling; the characteristic polynomial and spectrum (2) together with $S_1, \dots, S_4$ and s_1 at $\mathbf{1}$; the pairing identity (1) on several rational weights; every claim (1)–(4) about X , including the exact containment $0 \preceq X \preceq I$; and that $w^* \in \Delta_E$ with the stated spectrum and the two strict inequalities $\frac{27}{7} > 3$ and $\frac{99}{7} < 15$. Its recorded run reports 68 checks, all passing. It does not reprove the Ky Fan principle or Lemma 3, and it certifies nothing about minimality, about other graphs, or about other orders.

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